

Mojtaba Kalantari

RF & Microwave Engineer

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Iran, Tehran

Linkedin

Relevant Coursework

- Antenna Engineering
- Microwave Engineering
- Advanced Electromagnetics
- Electromagnetics of Complex Media
- Numerical Methods in Electromagnetics
- Metamaterials
- Active Microwave Circuits
- Electromagnetic Compatibility
- Optical Fiber

Skills

CST Studio Suite ●●●●●	Keysight ADS ●●●●●
Array Design ●●●●○	Antenna Design ●●●●○
MATLAB ●●●●○	Altium Designer ●●●●●
Mechanical design ●●○○○	RF Measurements ●●●●○
Python ●●○○○	Ansys HFSS ●●●●●
LNA Design ●●●●○	Filter Design ●●●●○
SolidWorks ●●●○○	FEKO ●●●●●

Languages

• Persian	• English
Native	B2(Improving)
• Deutsch	
A2 (Improving)	

RF & Microwave Engineer specializing in passive RF components, microwave filters, and antenna design for cellular communication systems. Experienced in electromagnetic simulation, RF hardware development, and prototype validation through both academic research and industrial projects. Interested in developing practical RF and microwave solutions for next-generation wireless communication systems.

Education

K. N. Toosi University of Technology Sep 2023 – Feb 2026

M.Sc. in Electrical Engineering, Telecommunication – Fields and Waves - 18.08/20 Ranked 1st

• Design and Simulation of a Diplexer in Two Frequency Bands 1710–1785 MHz and 1805–1880 MHz Using the Coupling Matrix Method 20/20

K. N. Toosi University of Technology Sep 2019 – Sep 2023

B.Sc. in Electrical Engineering, Telecommunication - 17.22 / 20 Ranked 1st

• Design and Construction of a Cross-Dipole Antenna with Reduced Interaction Effects on Nearby Antennas for Base Station Applications 20/20

Experience

WITEM Laboratory - R&D Mar 2023 - Present

Research coupling matrix synthesis of microwave filters and diplexers May 2025 - Dec 2026

• Designed practical mechanical diplexers with coaxial resonators for 1710–1785 MHz and 1805–1880 MHz using coupling matrix techniques

Design and develop BTS antenna Oct 2024 - Jul 2025

• Design, manufacture, and test electromagnetic cross-dipoles for 690–960 MHz, 1690–2690 MHz, and 2300–3800 MHz for 2G, 3G, 4G, and 5G mobile networks.

BTS Diplexer Jul 2024 - Aug 2024

• Design practical, low-cost microstrip diplexers with band-stop filters for 1690–2200 MHz and 2400–2690 MHz.

Phase Shifter Design Mar 2024 - Jul 2024

• Design and test a 9-element true-time-delay phase shifter for 690–960 MHz with 12° electrical tilt.

Fixed-Beam Array Oct 2023 - Jan 2024

• Design fixed-beam 12-element arrays for 2300–3800 MHz with 6° elevation.

Antenna Element Design Mar 2022 - Aug 2023

• Design, manufacture, and test parasitic elements for decoupling electromagnetic cross-dipole arrays in 690–960 MHz, and for tuning co-polarization and cross-polarization.

Teaching Assistant

Microwave Engineering Feb 2026 - Aug 2026

•Problem-Solving Sessions, Quiz & Assignment Grading ,ADS/CST Simulation Training, Practical Project Supervision - supervisor: Dr. Tavakol Pakizeh

Electromagnetic Fields and Waves Sep 2025 - Feb 2026

• Problem-Solving Sessions, Quiz & Assignment Grading, Electromagnetic Analysis Assistance, HFSS Simulation Training - supervisor: Dr. Seyed Arash Ahmadi

Antenna I & II Theory Sep 2023 - Aug 2024 & Feb 2024 - Feb 2025

• Problem-Solving Sessions, Quiz & Assignment Grading, Antenna Design Assistance, HFSS , CST, ADS Simulation Training - supervisor: Dr. Hadi Aliakbarian

Signals and Systems Feb 2023 - Aug 2023

• Problem-Solving Sessions, Quiz & Assignment Grading, Conceptual Learning Assistance - supervisor: Dr. Lotfollah Beygi

Design Logical Circuits Sep 2022 - Feb 2023

• Problem-Solving Sessions, Quiz & Assignment Grading, Digital Circuit Assistance - supervisor: Dr. Hamed Khanmirza

Certifications

Altium Designer Training Course - 44 Hours - Final Score: 88/100

Jan 2025 - Apr 2025

GARD

Publications

An improved printed Palm crossed dipole antenna with reduced interference on adjacent elements (Under Review)

Jul 2026

References

Hadi Aliakbarian

Associate Professor

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Website

Arash Ahmadi

Associate Professor

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Website

Tavakol Pakizeh

Associate Professor

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Website

Lotfollah Beygi

Assistant Professor

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Hamed Khanmirza

Assistant Professor

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Payaco Communication Industries

R&D Engineer Mar 2022 - Nov 2022

Troubleshooting, performance improvement, and optimization of testing and assembly processes.

RF & Microwave Technical Instructor Sep 2022 - Oct 2022

Conducted a six-session training course at Paya Co. covering BTS links, PIM, impedance matching, and radiation characteristics in cellular communication systems for assembly, testing, and troubleshooting personnel.

MOJPA - Internship Jun 2021 - Sep 2021

Antenna Engineering Intern

Design, manufacturing, and testing of a circular horn antenna with circular polarization covering the X-band using a turnstile OMT feed.

Projects

H.M.N - Repeater Design Apr 2023 – Present

• Designed & manufactured 2G-4G mobile repeaters in channel-selective and simple modes.

• Designed indoor and outdoor antenna elements for 2G-4G repeaters.

• Designed multilayer RF PCBs (1–4 layers), performed prototype assembly, RF measurements, troubleshooting, and hardware validation.

• Prepared engineering documentation and BOMs and supported prototype manufacturing and laboratory testing.

H.M.N - Absorber Manufacturing Apr 2023 – Present

• Manufactured and tested pyramidal absorbers for 1-12 GHz applications.

H.M.N - P2P Reflector Link Manufacturing Dec 2025 – Present

• Designed and tested hat-fed reflector antennas at 15, 18, 23, and 26 GHz with 2-, 3-, and 4-foot reflector sizes.

• Designed, manufactured, and tested absorbers for reflector antennas to reduce backlobe levels.

Metasurface Design for Enhanced MRI Image Quality May 2026- Aug 2026

• Designed and experimentally validated a metasurface operating at 63.8 MHz to enhance RF field distribution and improve MRI signal intensity and image quality.

IRIMO Sep 2024 – Nov 2024

• Designed, manufactured, and tested a 406 MHz omnidirectional antenna for weather balloon communication, providing coverage over 30°–70° elevation angles.

Microwave Filter & Diplexer Design Jul 2023 - Aug 2023

• Designed and optimized microwave filters for K-, Ku-, and Ka-band

Volunteer

IEEE Student Branch Volunteer 2023

• Organized MMWaTT Conference Feb 2023, ICEMD Conference & IEEE Student Ethics Competition Dec 2023

Student Scientific Organization of Electrical Engineering of Iran Sep 2021 - Sep 2022

• Secretary & Advisor, Regional Council Commission

• Specialized Seminar Organization, Student Association Coordination ,Educational Course Organization ,Scientific Program Planning ,Association Advisory Support ,Association Activity Evaluation ,University Coordination, Scientific Event Organization

Student Scientific Association of Electrical Engineering Sep 2020 - Sep 2022

• Core Member & Advisor

•Technical Course Organization, Freshman Welcome Events, Graduation Ceremony Organization